with the important case of its invariant distribution satisfying the partial differential equation

$$\hat{\mathcal{L}}^\dagger \hat{\rho} = 0.$$

Our goal in the following section is to find $\hat{\mathcal{L}}^\dagger$ and solve the PDE for the distribution $\hat{\rho}$ by using our ansatz (7.17). The solution gives us the corrupted invariant distribution sampled by our numerical scheme, allowing us to correct our observed averages to achieve more accurate results. It also gives us a way of evaluating which schemes are best for sampling, by evaluating the error introduced by the discretization.

The terms of the perturbation expansion for $\hat{\mathcal{L}}^\dagger$ can be computed using the Baker-Campbell-Hausdorff (BCH) expansion already introduced in Chap. 3. Recall that for linear operators X and Y, we can write the composition of their exponentials as

$$\exp(hX)\exp(hY) = \exp(hZ),$$

where we have the expansion

$$Z = X + Y + \frac{h}{2}[X, Y] + \frac{h^2}{12}([X, [X, Y]] + [Y, [Y, X]]) + \mathcal{O}(h^3).$$

The notation $[X, Y] \stackrel{\text{def}}{=} XY - YX$ represents the commutator of the operators X and Y. By repeated application of the BCH formula, we can obtain such an expansion for any splitting-based method, however the terms grow in complexity at successive powers of h, requiring commutators of increasing length that make computation cumbersome. Additionally any careful mathematician should question the convergence of this sequence when computing $\hat{\mathcal{L}}$, as we deal only with formal series that offer no guidance on the boundedness of the resulting terms, which will involve high order derivatives in both $\boldsymbol{q}$ and $\boldsymbol{p}$.

To offer some reassurance, let us define the evolution operator for our numerical method as

$$\mathcal{P}_t = e^{t\hat{\mathcal{L}}} = e^{t\mathcal{L}_1} e^{t\mathcal{L}_2} \dots e^{t\mathcal{L}_k},$$

where the numerical method has k-many stages with each $\mathcal{L}_i$ proportional to one of $\mathcal{L}_\mathrm{A}$, $\mathcal{L}_\mathrm{B}$ and $\mathcal{L}_\mathrm{O}$. We can make use of Taylor's theorem to write the identity

$$\mathcal{P}_h = \mathcal{P}_0 + h \left.\frac{\mathrm{d}\mathcal{P}_t}{\mathrm{d}t}\right|_{t=0} + \dots + \frac{h^n}{n!} \left.\frac{\mathrm{d}^n\mathcal{P}_t}{\mathrm{d}t^n}\right|_{t=0} + \frac{h^{n+1}}{n!} \int_0^1 (1-\lambda)^n \left.\frac{\mathrm{d}^{n+1}\mathcal{P}_t}{\mathrm{d}t^{n+1}}\right|_{t=\lambda h} \mathrm{d}\lambda,$$

for some finite truncation index $n \in \{1, 2, 3, \ldots\}$. The remainder term at order h^{n+1} will involve the integration of finitely many derivatives of its argument. Hence for observables of sufficient smoothness with compact support, this remainder term will remain bounded (see [223]).

7.7.2 First Order Operator Expansion

As we consider schemes involving compositions of three splitting pieces, the minimum number of stages to our methods (in order for it to be consistent) will be three. The perturbed Fokker-Planck operator for a splitting method with three stages will be computed as

$$\exp(hS) = \exp(hZ)\exp(hY)\exp(hX), \tag{7.18}$$

where X, Y and Z are one of $\mathcal{L}_{\mathrm{A}}^{\dagger}$, $\mathcal{L}_{\mathrm{B}}^{\dagger}$ and $\mathcal{L}_{\mathrm{O}}^{\dagger}$. Let us work the first terms of this expansion out, retaining terms involving triple products only

$$\exp(hY)\exp(hX) = \exp\left(h\left((Y+X) + \frac{h}{2}[Y,X] + \frac{h^2}{12}([Y,Y,X] + [X,X,Y]) + \ldots\right)\right),$$

where the notation $[A, B, C]$ is notation for $[A, [B, C]]$. Applying this formula one more time in (7.18), we find

$$\begin{aligned}
S &= X + Y + Z + \frac{h}{2}[Y,X] + \frac{h^2}{12}([Y,[Y,X]] + [X,[X,Y]]) \\
&\quad + \frac{h}{2}\left[Z, Y + X + \frac{h}{2}[Y,X]\right] \\
&\quad + \frac{h^2}{12}([Z,[Z,Y+X]] - [Y+X,[Z,Y+X]]) + \ldots \\
&= X + Y + Z + \frac{h}{2}([Z,Y] + [Z,X] + [Y,X]) + \frac{h^2}{4}[Z,Y,X] \\
&\quad + \frac{h^2}{12}([Y,Y,X] + [X,X,Y] + [Z,Z,Y] + [Z,Z,X] \\
&\quad + [Y+X, Y+X, Z]) + \ldots
\end{aligned}$$

In general the terms in this expansion will lead to complicated expansions. In the case of deterministic systems, the operators are always first order due to equality of mixed partials (the 2nd order derivatives are cancelled). Due to the second derivatives present in the $\mathcal{L}_{\mathrm{O}}^{\dagger}$ operator, derivatives of all orders will appear in the